Rectifiers, Zener regulators, clippers

Cheat sheet · Day 05 · Wed 28 Oct · pack E3

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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RECTIFIERS   Vm = √2·Vs(rms).  Vp = Vm − VD (HW, CT)  |  Vm − 2VD (bridge)
  Vdc(no C): HW Vm/π   FW 2Vm/π    f_out: HW f, FW 2f
  PIV: HW Vs (≈2Vm WITH C) | CT 2Vs − VD | bridge Vs − VD ; rate ≥1.5×PIV
  With rf: Im = Vm/(rf+R) [bridge 2rf+R]; HW Idc=Im/π, Irms=Im/2; FW 2Im/π, Im/√2
FILTER (CR >> T)   Vr(pp) = Vp/(fCR) = IL/(fC)   FULL-WAVE: use 2f
  Design: C = Vp/(f R Vr) [or 2f].  Vo(dc) = Vp − Vr/2,  IL = Vo/R
  ωΔt = √(2Vr/Vp) rad ; fraction = ωΔt/2π ; Δt = ωΔt/(2πf)
  HW: iDav = IL(1+π√(2Vp/Vr))   iDmax = IL(1+2π√(2Vp/Vr))
  FW: iDav = IL(1+π√(Vp/2Vr))   iDmax = IL(1+2π√(Vp/2Vr))
  Ripple factor r = Vr(rms)/Vdc, Vr(rms) = Vr(pp)/(2√3) -> 10% => Vr(pp)=0.346 Vdc
  %VR = (VNL − VFL)/VFL × 100
ZENER   VZ = VZ0 + rz·IZ ;  IS = (Vs−VO)/RS = IZ + IL
  Analyse: (VO−VZ0)/rz + VO/RL = (Vs−VO)/RS ; check IZ>0 ; PZ = VO·IZ
  Design: VZ = VL ; RS = (Vs,min − VZ)/(IL,max + IZ,min)   [PT Q4: 20 Ω]
          PZ,max = VZ[(Vs,max − VZ)/RS − IL,min]  (no load!)  [PT Q4: 2.4 W]
  Load range: RL,min = VZ/(IR,min − IZ,min) ; RL,max = VZ/(IR,max − IZ,max) (∞ if ≤0)
  Line reg = (rz||RL)/(RS + rz||RL) ; Load reg = −(rz||RS)
  Filtered supply: use the ripple VALLEY Vp − Vr as Vs,min
CLIPPER: assume OFF -> vd(vi) -> set vd = VD -> viT -> vo per region -> sketch under
  [PT Q5: +20/−5 V square, 4 V battery opposing: vo = 0 / −1 V]
CLAMPER: C charges to Vm, waveform shifts by Vm, same p-p. DOUBLER: ≈2Vm, PIV 2Vm
LED λ(nm) ≈ 1240/Eg(eV); Schottky low VF, fast; varactor Cj↓ as VR↑; photodiode reve
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